

AAYUSH TIWARI

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SUMMARY

Java Backend Engineer experienced in designing and building REST APIs with Spring Boot, Spring Security, and Spring Data JPA. Delivered three end-to-end systems covering API design, JWT-based authentication, relational database modeling, third-party API integration, and containerized deployment. Strong DSA foundation with 200+ problems solved on LeetCode.

SKILLS

Languages: Java, Python, JavaScript, TypeScript

Backend: Spring Boot, Spring Data JPA, JWT, REST APIs, FastAPI

Frontend: React, HTML, CSS

Database: MySQL, PostgreSQL

AI/ML: scikit-learn, Random Forest, Pandas, NumPy, Streamlit

Blockchain: Solidity, Hardhat, Web3.js, Smart Contracts, DeFi

Tools: Git, GitHub, Docker, Maven, Postman, IntelliJ IDEA, VS Code

EXPERIENCE

Software Engineer Intern | Accumet Solutions

Jun 2025 – Dec 2025

- **Built core modules for a Meter Data Management System (MDMS)** that supported daily testing and validation of 300+ smart electrical 3-phase and water meters.
- **Developed data-handling logic for utility metering hardware**, streamlining real-time collection and processing of meter readings for downstream monitoring dashboards.
- **Collaborated with senior engineers to test, debug, and refine MDMS components**, improving data reliability and consistency across the daily meter-testing workflow.

PROJECTS

HipHopHub | Full-Stack Web Application | [Live Demo](#) | [GitHub](#)

Jan 2026 – May 2026

React, TypeScript, Spring Boot, Java, SQL, REST APIs, Spring Security, JWT, Docker

- **Deployed a production-ready full-stack platform** with a Dockerized backend on Render, a React frontend on Vercel, a layered Spring Boot REST API, JWT auth, and Last.fm/iTunes API integration.
- **Improved catalog data quality across 75 artists** by removing 93 contaminated tracks and 102 invalid releases through an automated multiphase classification pipeline for remix, compilation, and duplicate-artist contamination.
- **Increased session depth beyond passive browsing** with an interactive 'Guess the Song' game: configurable snippet durations (1–30s), quizzes, a global leaderboard, and achievement tracking.

Sentinel AI | AI-Powered Behavioral Anomaly Detection for Cybersecurity | [Live Demo](#) | [GitHub](#)

Mar 2026 – Jun 2026

Python, scikit-learn, FastAPI, Streamlit, Isolation Forest, Random Forest, SHAP, Pandas, NumPy, Plotly

- **Trained on 150K synthetic security events across 10 attack classes** to build a hybrid SOC prototype combining anomaly detection, attack classification, UEBA baselining, and a 0–100 risk engine with SHAP explainability.
- **Achieved a 0.9292 weighted F1 score and 91.85% accuracy** on held-out evaluation data, then shipped a FastAPI backend to Render and a Streamlit analyst dashboard to Streamlit Community Cloud.
- **Replaced a single opaque score with per-class reliability** via a six-module analyst workflow: Threat Intel, Incident Investigation, Entity Intel, Live Analyzer, Attack Story, and AI Trust Center.

ACHIEVEMENTS

- **Google Summer of Code (GSoc)** — Accepted as an open-source contributor; delivered code changes reviewed and merged by project maintainers.
- **Flipkart Gridlock** — Participated in Flipkart's latest Gridlock engineering challenge, competing against top B.Tech students across India.
- **200+ LeetCode Problems Solved** — Solved 200+ problems across arrays, trees, graphs, DP, and system design topics, building a strong foundation in data structures and algorithms.
- **450+ GitHub Contributions** — Maintained an active open-source contribution graph with 450+ commits across personal projects and collaborative repositories.
- **Bharat Academix Hackathon & Mahakumbh Tech Challenge** — Designed and presented technical solutions in two national-level hackathons under time-constrained conditions.

EDUCATION

B.Tech, Computer Science & Engineering

2023 – 2027

Vellore Institute of Technology, Bhopal · CGPA: 8.27 / 10

CERTIFICATIONS

- Google IT Support Professional Certificate — Coursera
- Bits and Bytes of Computer Networking — Coursera
- AI & Robotics Workshop — Techkriti, IIT Kanpur
- Introduction to Machine Learning — NPTEL
- Marketing Analytics — NPTEL